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B.Tech. CSE (Data Science and Analytics) (DSA)

SEMESTER V

LAB

PROJECT REPORT

AI PULSE: ANALYSIS OF AI TOOL ADOPTION AND ITS IMPACT ON STUDENTS

Submitted by

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DST Lab Project Report

1. PROJECT OBJECTIVE

The objective of AI Pulse is to explore and communicate how students use artificial intelligence (AI) tools in an academic context. The project uses data visualization in Tableau to summarize patterns in AI-tool preference, adoption across academic streams, student perceptions, and reported academic impact.

Load and explore the selected student AI-usage dataset.

Understand the available fields and distinguish categorical dimensions from numerical measures.

Prepare the data for visualization by checking data types, missing values and duplicate records, and handling issues only where required.

Create clear Tableau worksheets and combine them into four interactive dashboards.

Present findings in a form that helps viewers compare AI-tool usage and student perceptions across groups.

2. INTRODUCTION

Artificial intelligence tools are increasingly used by students for learning-related activities such as understanding concepts, generating ideas, summarizing information and supporting academic work. However, adoption and perceptions may differ across academic streams and individual students.

AI Pulse is a data-visualization project focused on exploring these patterns. Tableau is used to convert the dataset into charts and dashboards so that comparisons can be made more easily than by reading rows of raw data. The dashboards are descriptive: they summarize the responses available in the dataset and do not establish that AI tools caused a particular academic outcome.

2.1 Problem Statement

The project addresses the difficulty of understanding student AI-tool usage and perceptions from raw survey-style data. It aims to organize the available responses into visual summaries that show which tools are preferred, how patterns vary across academic streams, and how students report trust and academic impact. The project does not claim to measure the effectiveness of AI tools experimentally.

3. PROJECT METHODOLOGY

The project follows a data-visualization workflow: dataset identification, loading, exploration, basic data-quality checks, worksheet creation, dashboard assembly, and interpretation of the displayed results.

3.1 Dataset Loading and Exploration

Dataset name: Students(1).csv — student AI tool usage dataset (uploaded source file).

Dataset source/link: The dataset was supplied as Students(1).csv. An original public download URL or publisher was not included with the file.

Dataset structure: The supplied CSV contains 3,614 records (rows) and 16 fields (columns). It has 1,614 missing values, all in the State field, and 0 fully duplicated rows.

Dataset detail	Value to enter from the actual file
Dataset file	Students(1).csv (uploaded source file)
Number of rows / records	3,614
Number of columns / fields	16
Column names	Student_Name, College_Name, Stream, Year_of_Study, AI_Tools_Used, Daily_Usage_Hours, Use_Cases, Trust_in_AI_Tools, Impact_on_Grades, Do_Professors_Allow_Use, Preferred_AI_Tool, Awareness_Level, Willing_to_Pay_for_Access, State, Device_Used, Internet_Access
Data types	Integer: Year_of_Study, Trust_in_AI_Tools, Impact_on_Grades, Awareness_Level; Decimal: Daily_Usage_Hours; Text/category: Student_Name, College_Name, Stream, AI_Tools_Used, Use_Cases, Do_Professors_Allow_Use, Preferred_AI_Tool, Willing_to_Pay_for_Access, State, Device_Used, Internet_Access.
Missing values	1,614 missing values in total; all 1,614 missing values are in State. Other columns have no missing values.
Duplicate rows	0
Unit of observation	One dataset record per student response, based on the structure and field names; verify this against the dataset documentation if available.

Column / field	CSV data type	What it represents (based on field name)
Student_Name	object	Categorical or text

		response; exact categories depend on the dataset values
College_Name	object	Categorical or text response; exact categories depend on the dataset values
Stream	object	Categorical or text response; exact categories depend on the dataset values
Year_of_Study	int64	Student's year of study
AI_Tools_Used	object	Categorical or text response; exact categories depend on the dataset values
Daily_Usage_Hours	float64	Daily duration of AI-tool use, in hours
Use_Cases	object	Categorical or text response; exact categories depend on the dataset values
Trust_in_AI_Tools	int64	Numerical survey response / rating; confirm scale coding from questionnaire
Impact_on_Grades	int64	Numerical survey response / rating; confirm scale coding from questionnaire
Do_Professors_Allow_Use	object	Categorical or text response; exact categories depend on the dataset values
Preferred_AI_Tool	object	Categorical or text response; exact categories depend on the dataset values
Awareness_Level	int64	Numerical survey response / rating; confirm scale

		coding from questionnaire
Willing_to_Pay_for_Access	object	Categorical or text response; exact categories depend on the dataset values
State	object	Categorical or text response; exact categories depend on the dataset values
Device_Used	object	Categorical or text response; exact categories depend on the dataset values
Internet_Access	object	Categorical or text response; exact categories depend on the dataset values

Exploration steps: The CSV was read and its dimensions, field names, data types, missing values and duplicate rows were checked. The dataset contains 3,614 rows and 16 columns. State contains 1,614 blank values; the other 15 columns contain no missing values. No fully duplicated rows were found.

3.2 Data Preprocessing

Only preprocessing that was actually performed should be reported. Before final submission, confirm whether the source file contained null values or duplicate records and whether any corrections were made.

The uploaded CSV contains 1,614 missing values, all in State. The other 15 columns have no missing values. The report records this condition; do not claim the missing State values were filled or removed unless that was actually done.

The uploaded CSV contains 0 fully duplicated rows.

Verify that categories such as academic stream and AI-tool names are consistently spelled.

Confirm that numerical fields are stored as numeric values and categories as dimensions.

Do not replace missing responses with assumed values. If missing values were retained or excluded, state that choice explicitly.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version	Purpose
Tableau Desktop	2026.2.3 (20262.26.0912.1023)	Data connection, data exploration, calculated fields, worksheets, filters and dashboards.
Microsoft Excel / CSV	–	Dataset storage and basic inspection, if used.
GitHub	Web platform	Version-controlled project repository and sharing of the Tableau workbook, report and permitted dataset/link.

4.2 Step-wise Implementation

Open Tableau Desktop and connect to the supplied dataset (CSV / Excel file).

Review the Data Source page and verify field names, data types and the number of records.

Open individual worksheets and place categorical fields on Rows or Columns and measures on the Marks card as appropriate.

Choose chart types based on the question: comparison charts for category comparisons, heatmaps for two-dimensional category comparisons, and count-based views for response distributions.

Use Tableau aggregations such as COUNT or COUNTD only when they match the intended unit of analysis. Confirm whether the measure counts records or unique students.

Add labels, titles, legends, tooltips and filters so that the charts can be interpreted.

Create four dashboards and arrange the relevant worksheets. Check sizing, alignment, background formatting and readability.

Test filters and dashboard interactions, then save the final Tableau workbook (.twb or packaged .twbx, according to the submission requirement).

Capture screenshots of the final dashboards and add them to the Results and Discussion section.

5. DAX QUERIES / CALCULATED FIELDS

DAX is used in Power BI. Since this project is implemented in Tableau, the relevant logic is expressed through Tableau aggregations and calculated fields rather than DAX. Include only calculations that exist in the submitted workbook.

Calculation / aggregation	Purpose / explanation
COUNT([Student / Response ID])	Counts non-null records in the selected ID field. Use only if this is the actual field used in the workbook.
COUNTD([Student / Response ID])	Counts distinct IDs and can represent unique students if the ID field uniquely identifies each student.
Number of records / student count	A count measure used to display the number of responses or students; confirm which one the workbook represents.
AVG([Numeric survey field])	Calculates the average of a numeric response field, if an average was used in a dashboard.

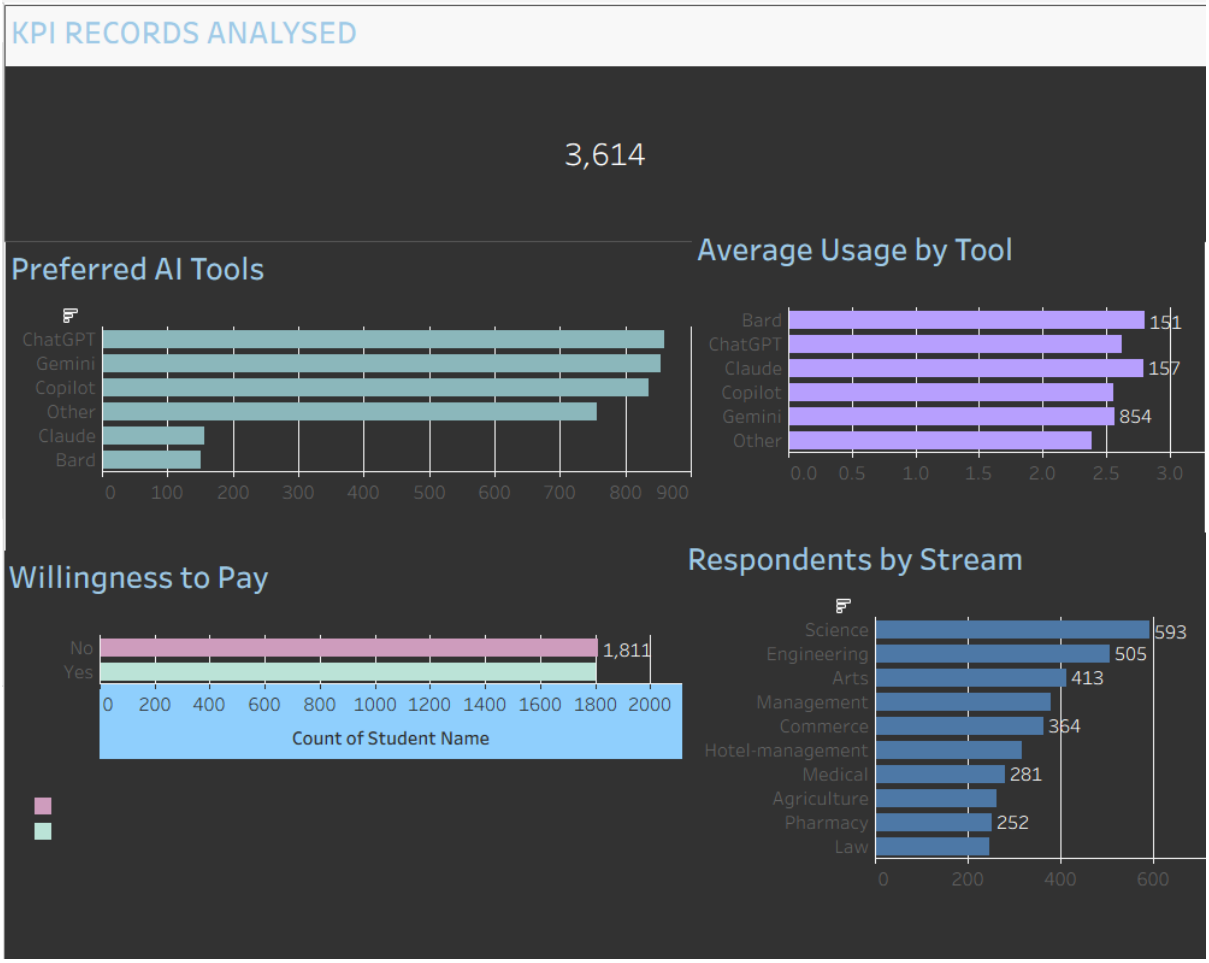
Important: The expressions above are examples of Tableau calculation patterns, not confirmation that each one exists in the workbook. Replace the field names with the exact names and remove any calculation that was not implemented. If no custom calculated fields were created, write: “No custom calculated fields were required; the visualizations use Tableau’s built-in aggregations.”

6. RESULTS AND DISCUSSION

The project produces four Tableau dashboards. Insert the actual screenshots exported from the final workbook in the spaces below. Each figure should be readable and should match the final submitted workbook.

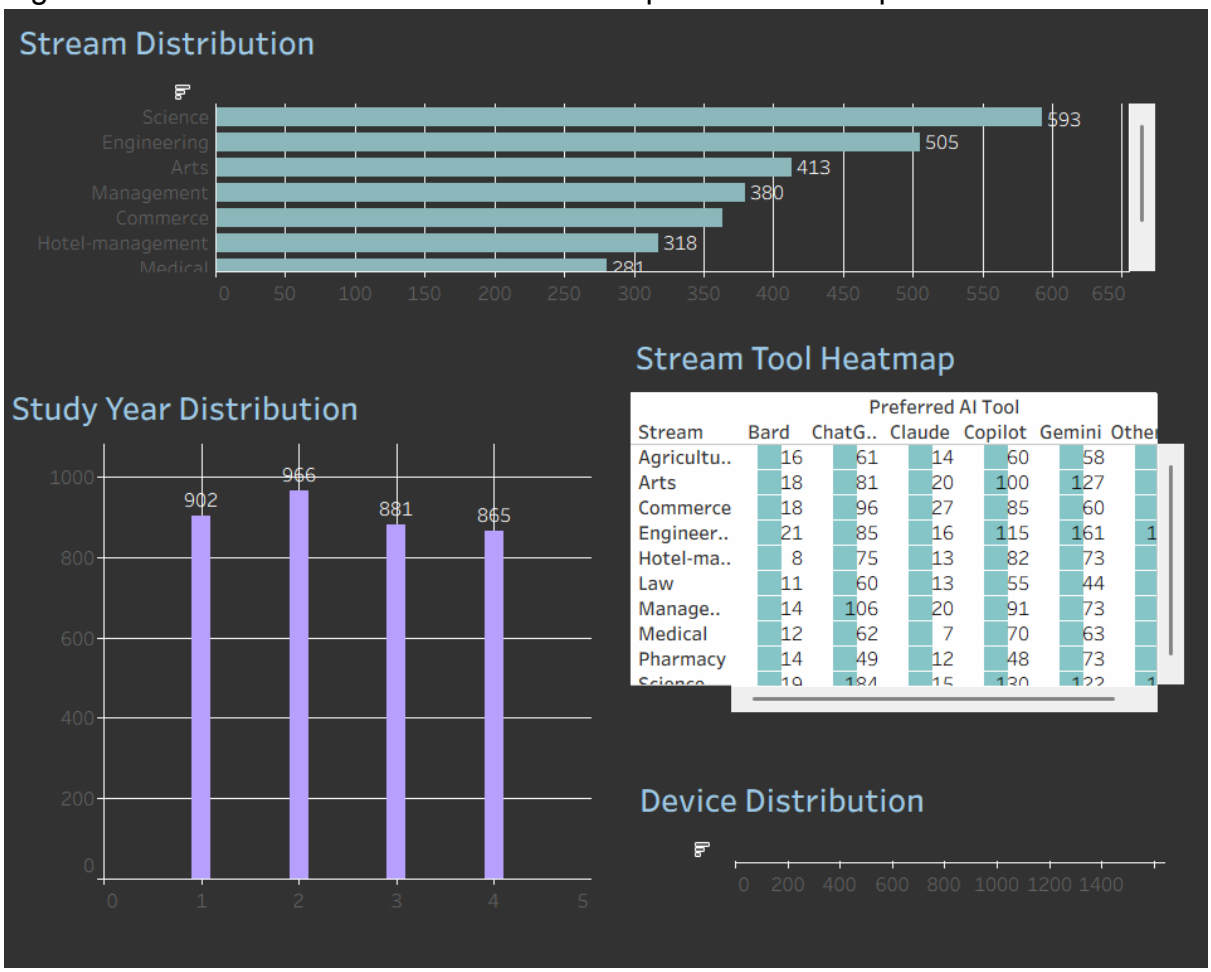
Dashboard 1 – AI adoption overview

Figure 1. Dashboard 1: AI adoption overview.



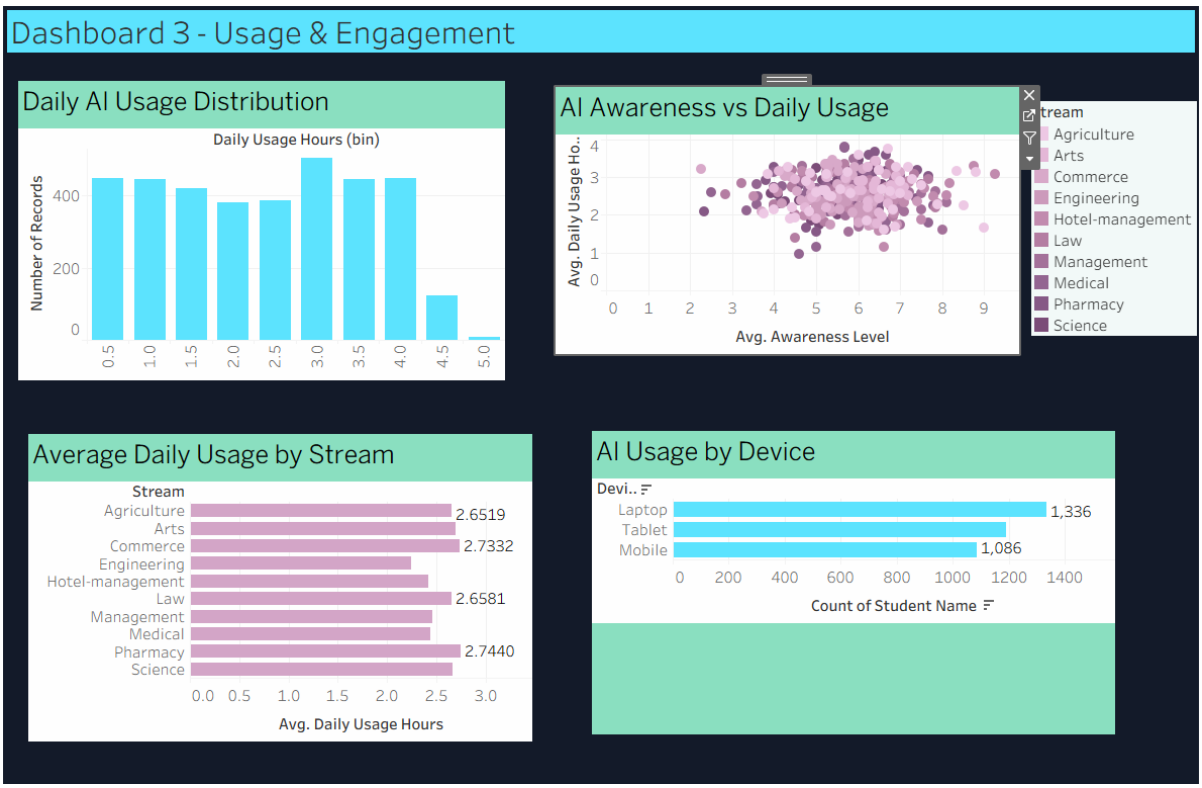
Dashboard 2 – Student profile and preferred AI tools

Figure 2. Dashboard 2: Student profile and preferred AI tools.



Dashboard 3 – Student usage and impact analysis

Figure 3. Dashboard 3: Student usage and impact analysis.



Dashboard 4 – Trust and academic perceptions

Figure 4. Dashboard 4: Trust and academic perceptions.



Overall interpretation

Together, the dashboards provide a visual summary of AI-tool adoption and student perceptions in the dataset. They support quick comparisons between categories and make response patterns easier to communicate. Findings should be limited to the population represented by the dataset and the questions actually asked in the source survey.

7. CONCLUSION

AI Pulse demonstrates how Tableau can be used to explore a student AI-usage dataset and communicate patterns through multiple dashboards. The project organizes the available responses into visual comparisons of AI-tool preference, usage and student perceptions. The work is descriptive and does not train a machine-learning model.

Limitations: conclusions depend on the quality and representativeness of the source dataset; survey responses may reflect self-reported perceptions; missing or inconsistent responses can affect counts; and descriptive visualizations cannot establish causation. The exact sampling method and dataset provenance should be stated only when confirmed from the original source.

Future improvements may include documenting the survey sampling method, adding more filters and clearer tooltips, validating calculated measures against the raw data, refreshing the dashboard with newer responses, and adding statistical analysis if supported by the research question and dataset.

8. GITHUB LINK

GitHub Repository URL: https://github.com/VanshikaRao21/consumer-ai-tool-engagement

Suggested repository contents: final Tableau workbook (.twb / .twbx), report (DOCX and final PDF), dashboard screenshots, README with project overview and setup instructions, and either the dataset (only if its licence permits redistribution) or the original dataset link.

9. REFERENCES

[1] Dataset-Kaggle

[2] Tableau, “Tableau Desktop Help,” https://help.tableau.com/current/pro/desktop/en-us/

[3] GitHub Docs, “About repositories,” https://docs.github.com/en/repositories